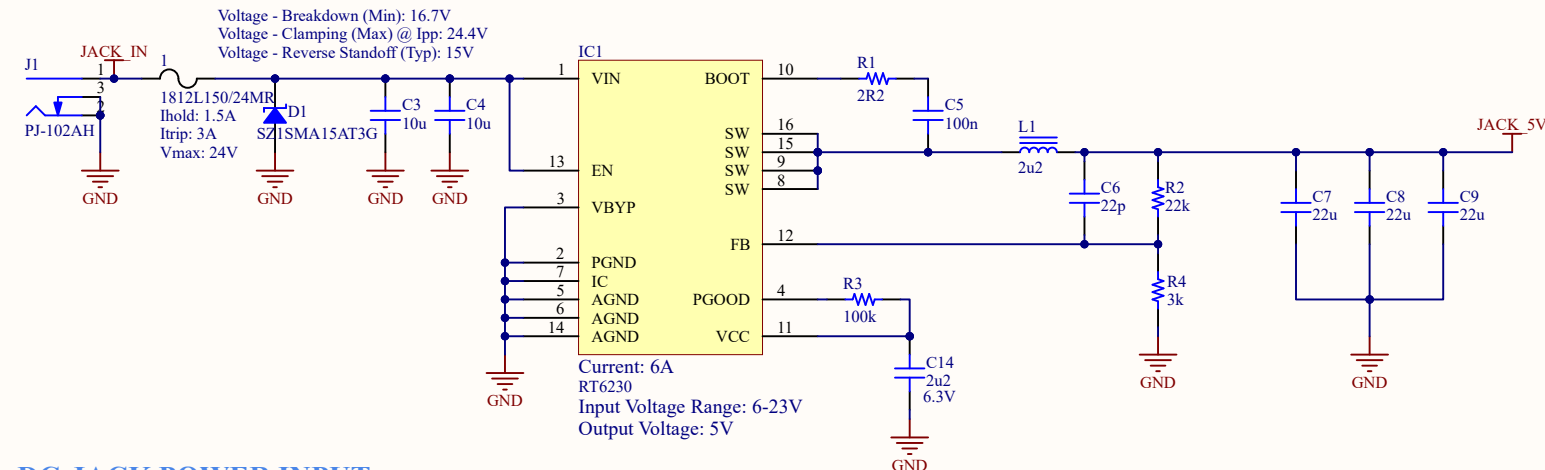


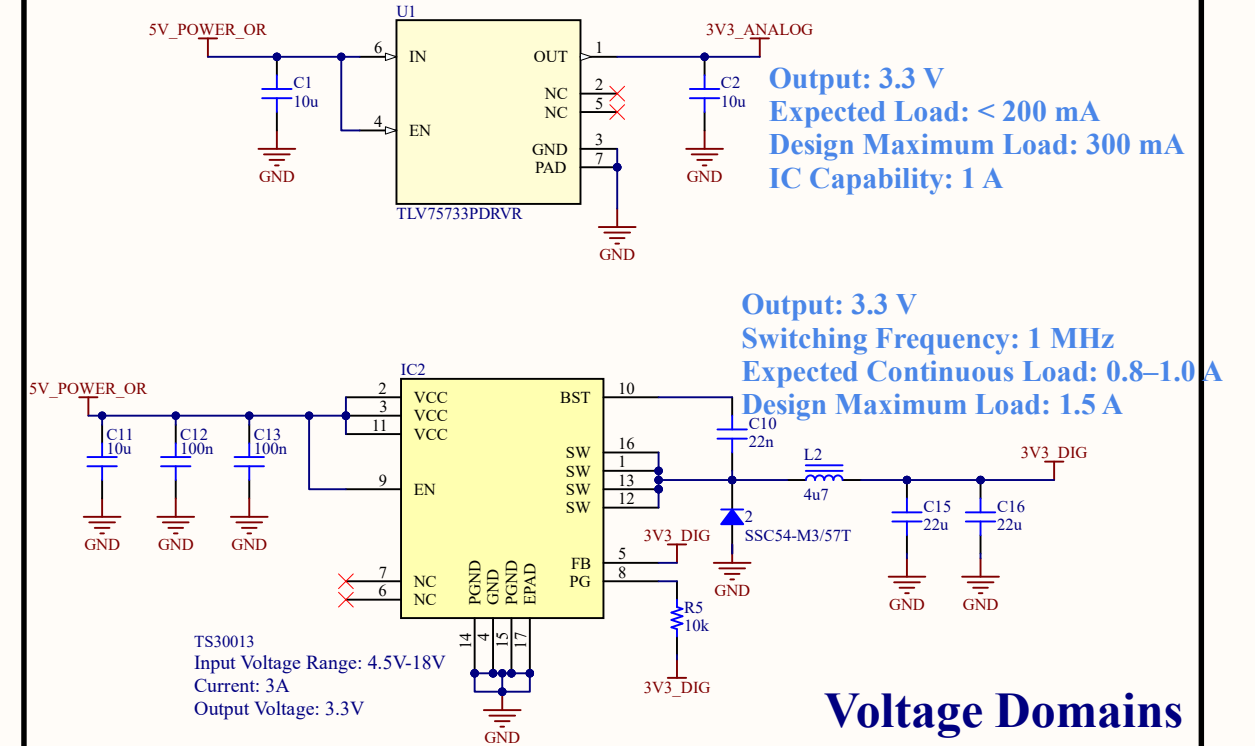
# Power Section



**DC-JACK POWER INPUT**  
**9–12 VDC REGULATED, CENTER-POSITIVE**  
**Recommended Adapter: 9 V / 2 A or 12 V / 1.5 A**  
**JACK\_5V: 5.0 V / 1.5 A CONTINUOUS**  
**2.0 A PEAK**

**USE CENTER-POSITIVE**  
**ADAPTER ONLY, NO**  
**REVERSE-POLARITY**  
**PROTECTION !!!**

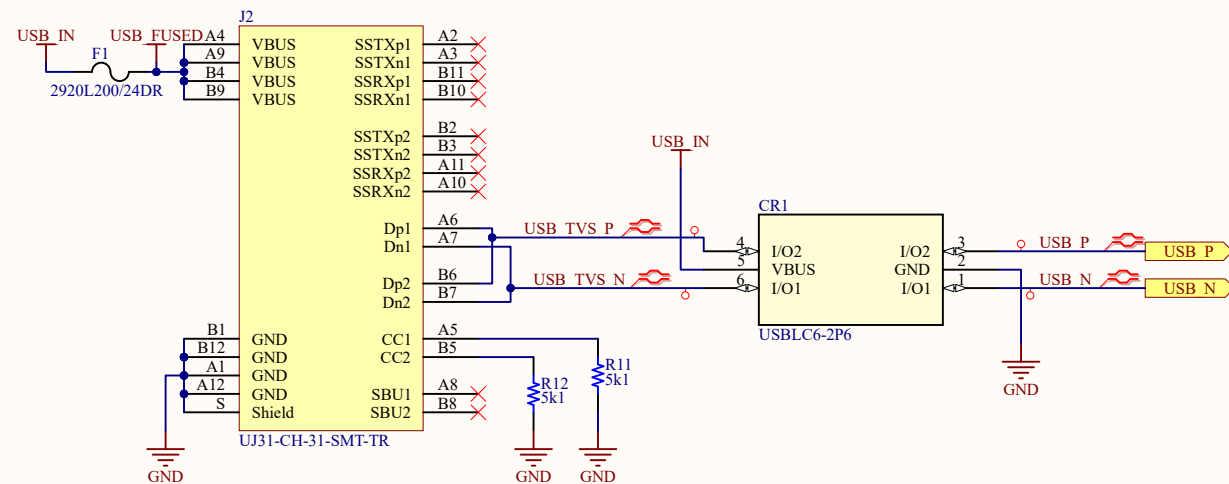
**DC-Jack Input**



**Output: 3.3 V**  
**Expected Load: < 200 mA**  
**Design Maximum Load: 300 mA**  
**IC Capability: 1 A**

**Output: 3.3 V**  
**Switching Frequency: 1 MHz**  
**Expected Continuous Load: 0.8–1.0 A**  
**Design Maximum Load: 1.5 A**

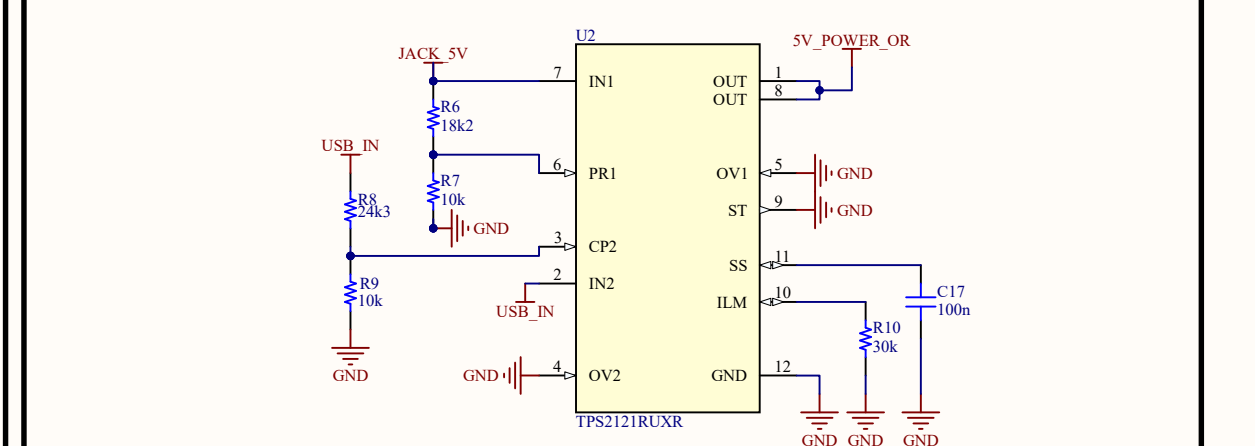
**Voltage Domains**



**RECOMMENDED ADAPTER: 5 V / 3 A**  
**TYPE-C SOURCE**  
**OR 5V/2A TYPE-A-LIKE SOURCE**

**USE A 5 V TYPE-C POWER SOURCE ONLY.**  
**NO USB POWER DELIVERY SUPPORT.**  
**DO NOT TRY HIGH-CURRENT OPERATION**  
**FROM AN UNKNOWN PC OR USB CHARGER.**

**USB-TYPE-C Input**

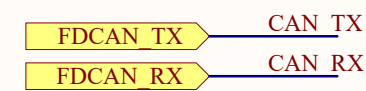


**The 9–12 V DC-jack input is converted to a regulated 5 V rail by the RT6230 buck converter. TPS2121 prioritizes JACK\_5V and automatically transfers the load to USB\_IN if the primary 5 V rail falls below the threshold which is 4.11V. The resulting 5V\_POWER\_OR rail supplies the analog and digital 3.3 V regulators.**

**Power MUX**

| Title |                              |           |
|-------|------------------------------|-----------|
| Size  | Number                       | Revision  |
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| File: | C:\Users\PowerNetwork\SchDoc | Drawn By: |

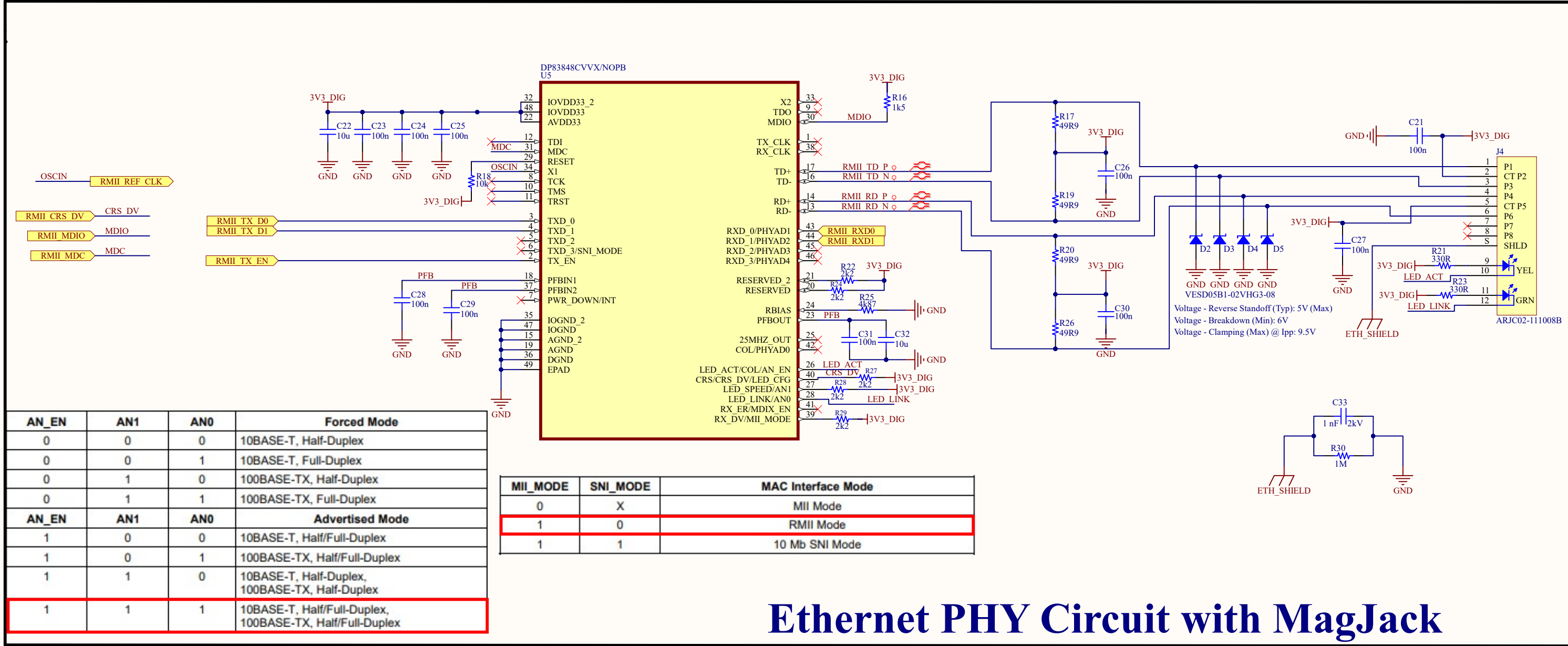
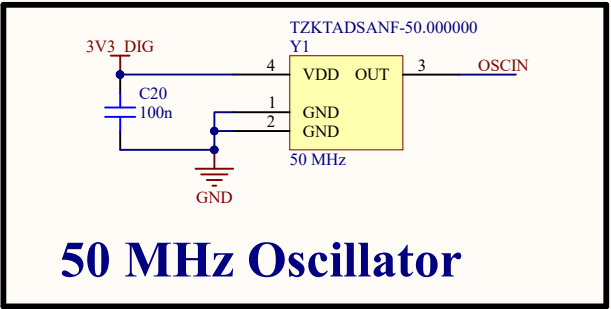
|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|



**CAN FD INTERFACE** — Use 120  $\Omega$  twisted-pair cable. The split termination should be populated **ONLY** when this board is located at a physical end of the bus. Intermediate nodes must be assembled without R14 and R15.

|            |                            |           |
|------------|----------------------------|-----------|
| Title      |                            |           |
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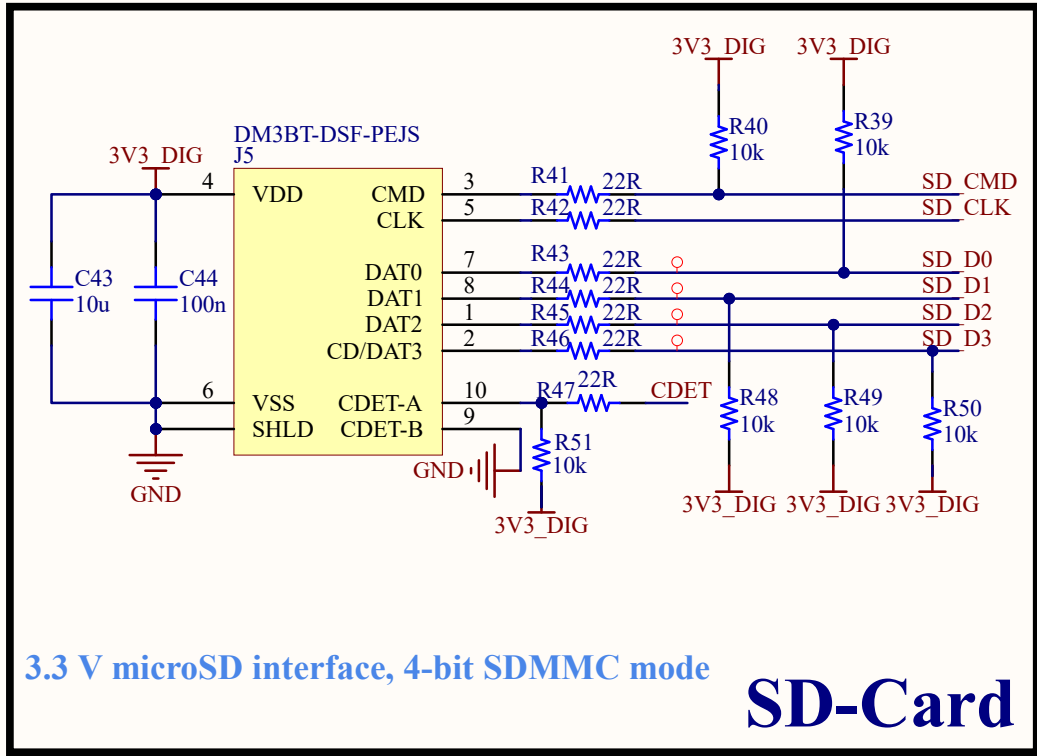
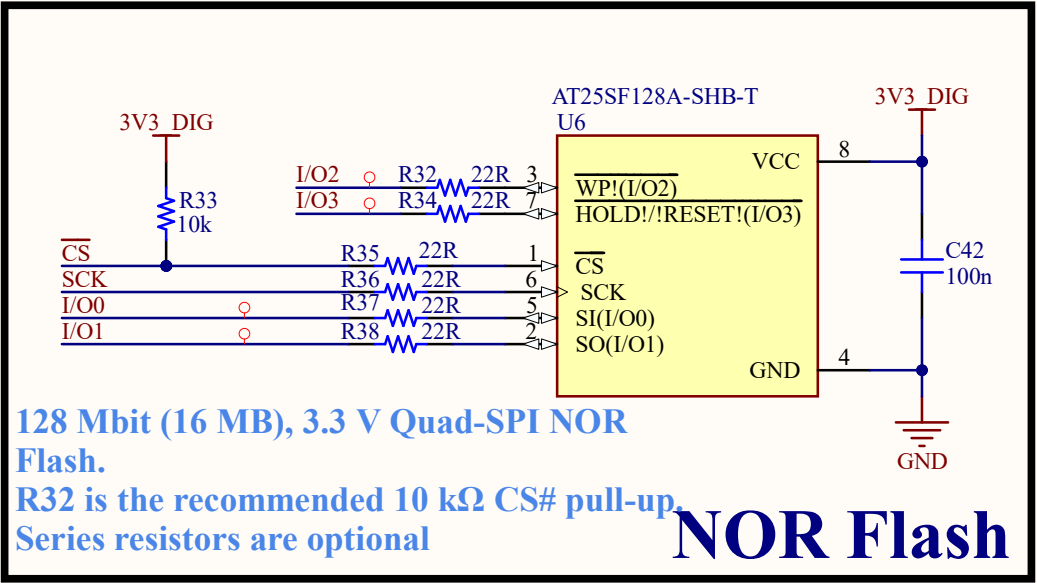
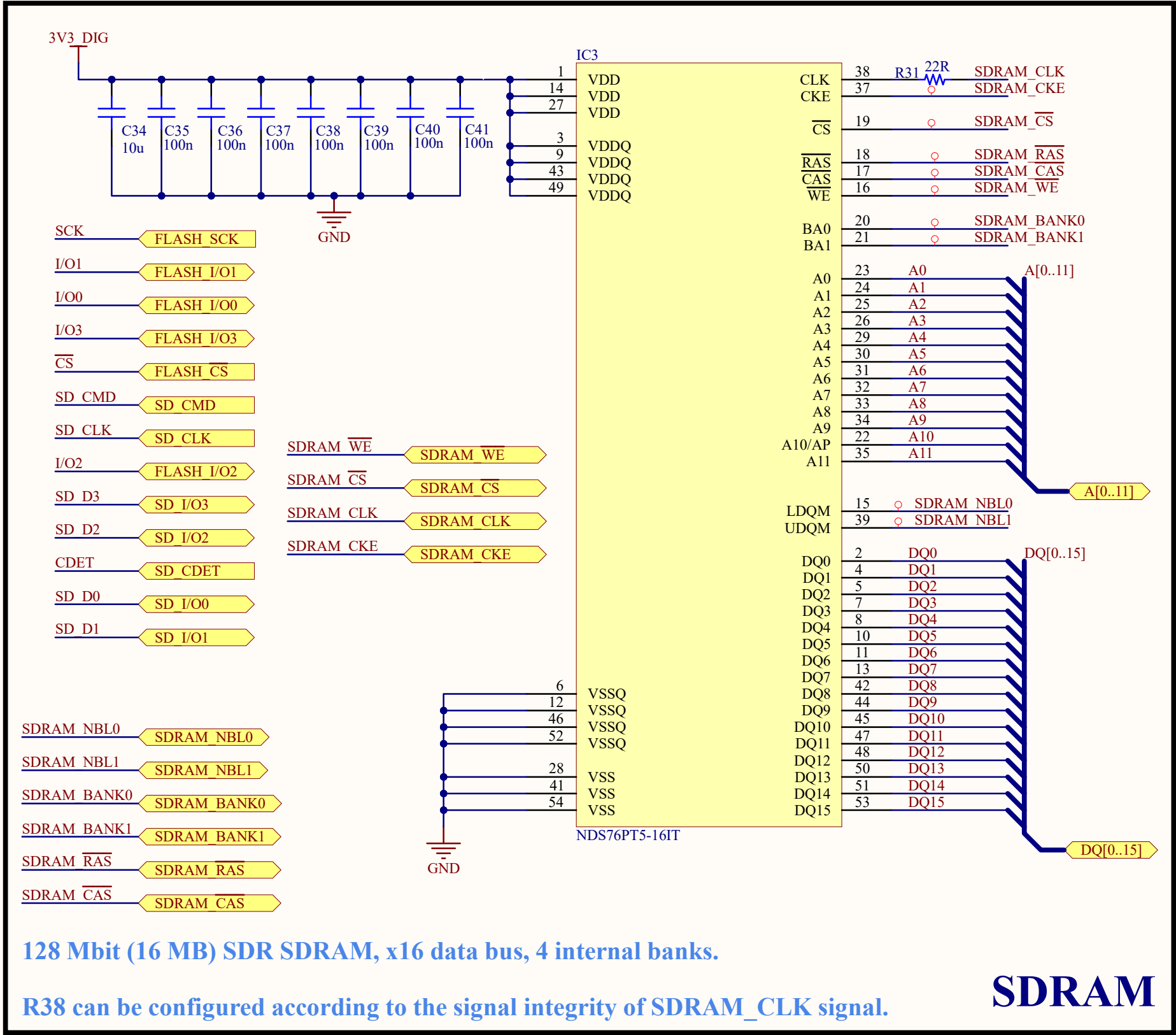
# Ethernet Section



## Ethernet PHY Circuit with MagJack

|       |                              |           |
|-------|------------------------------|-----------|
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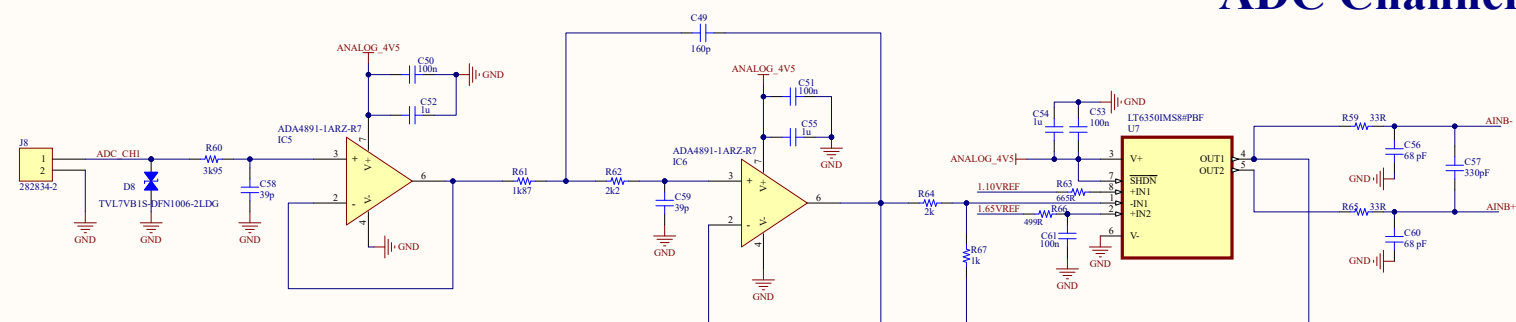
# Memory Section



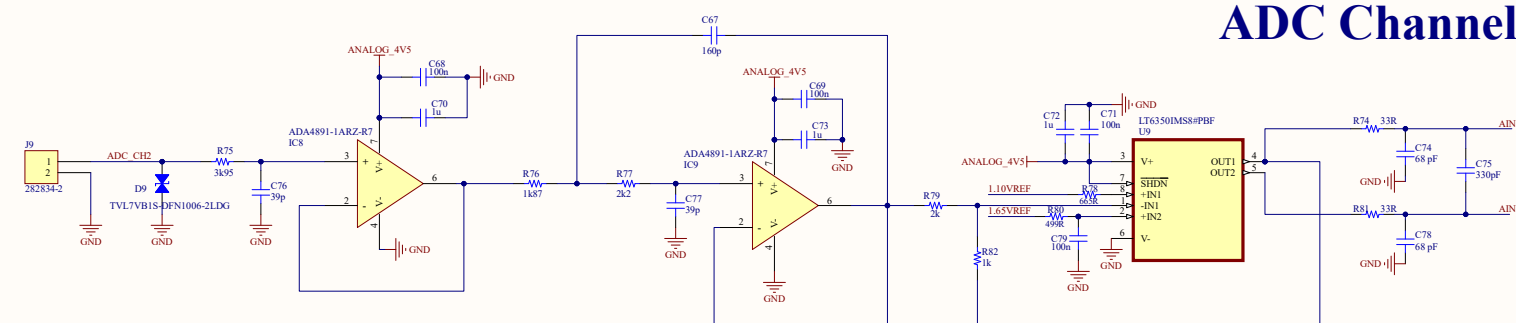
| Title |                            |           |
|-------|----------------------------|-----------|
| Size  | Number                     | Revision  |
| A4    |                            |           |
| Date: | 9.02.2026                  | Sheet of  |
| File: | C:\Users\...\Memory.SchDoc | Drawn By: |

# Analog Circuitry

ADC Channel 1



ADC Channel 2



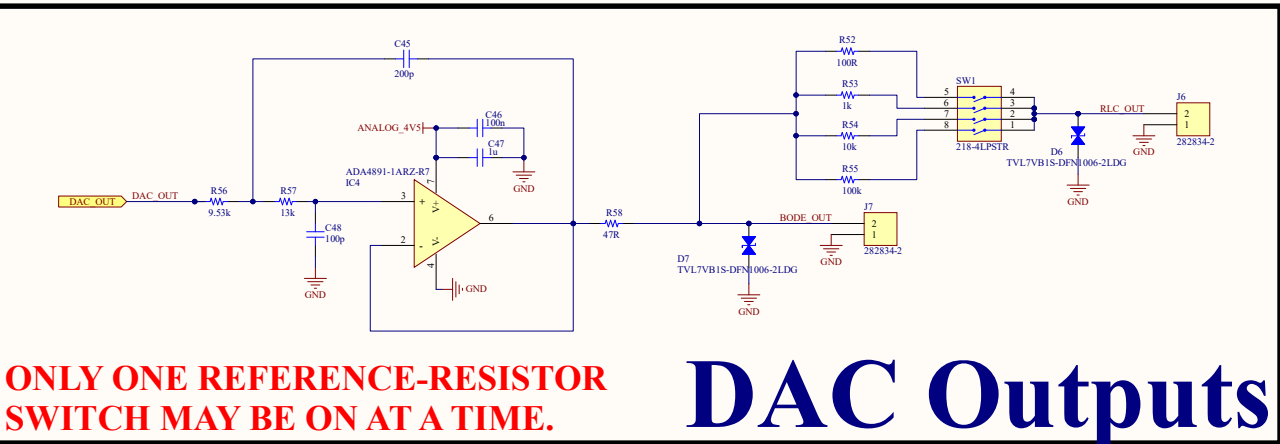
Input RC pole  $\approx 1.03$  MHz; second-order stage cutoff frequency  $\approx 994$  kHz, together forming a 3rd-order Butterworth response, overall  $-3$ dB bandwidth  $\approx 1.0$  MHz. ADC input network:  $33\ \Omega$  /  $68\ \text{pF}$  /  $330\ \text{pF}$ .

Do not apply negative voltage or signals above 3.3 V.

## ADC Inputs

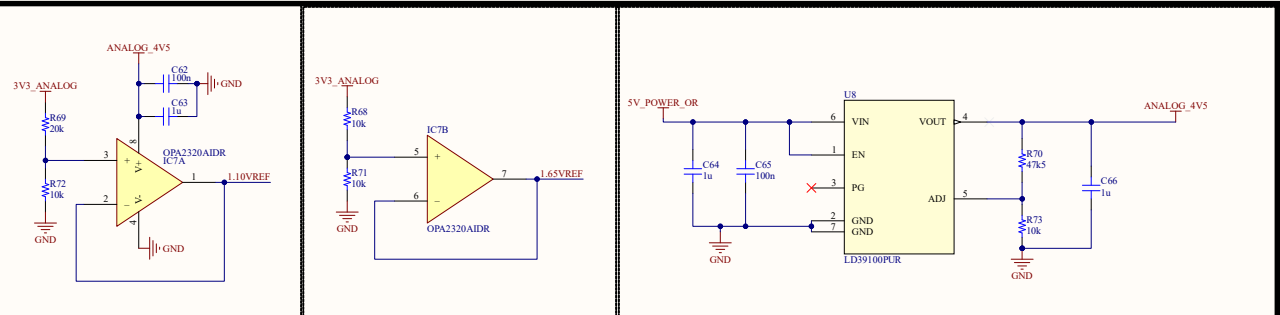
Two simultaneously sampled, 16-bit channels operate at up to 4 MSPS. Each external single-ended input is protected, buffered, low-pass filtered, level-shifted, and converted to a differential signal centered at 1.65 V before driving the AD7380. The ADC input networks use the manufacturer-recommended  $33\ \Omega$  /  $68\ \text{pF}$  /  $330\ \text{pF}$  differential RC configuration. The nominal external input range is 0 V to 3.3 V referenced to board GND.

The integrated DAC signal path provides programmable excitation for frequency-response and impedance measurements. BODE\_OUT is the filtered excitation output; RLC\_OUT is the node after the manually selected reference resistor. Select only one reference resistor at a time. External test equipment and the device under test must share board GND. The analog interfaces are not isolated and are not permanently  $50\ \Omega$  terminated.



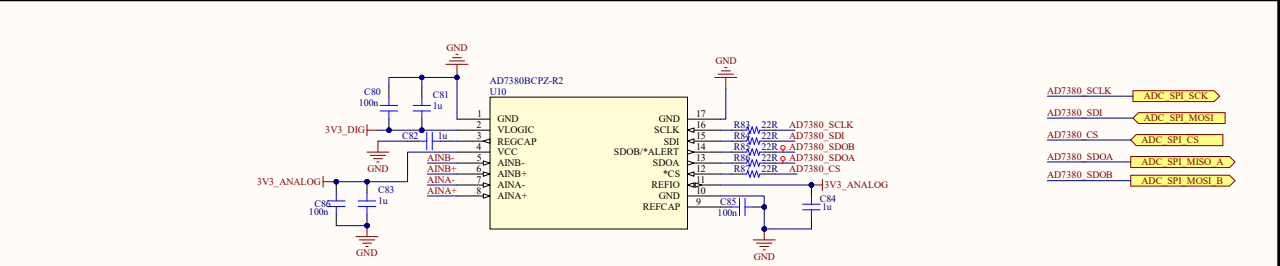
ONLY ONE REFERENCE-RESISTOR SWITCH MAY BE ON AT A TIME.

## DAC Outputs



The adjustable LD39100 generates a nominal 4.5 V analog rail from 5V\_POWER\_OR. Buffered 1.10 V and 1.65 V bias references are derived from 3V3\_ANALOG using an OPA2320. The 1.65 V node defines the differential-driver output common-mode voltage, while the 1.10 V node provides the input level-shift reference.

## Voltage References



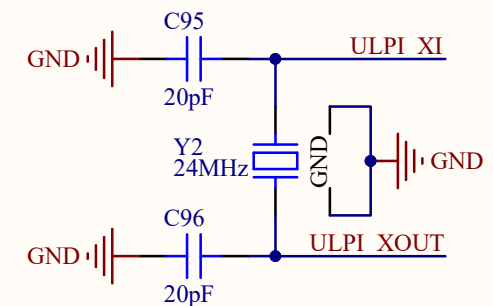
Dual simultaneous-sampling, 16-bit SAR ADC operating at up to 4 MSPS per channel. VCC is supplied from 3V3\_ANALOG and VLOGIC from 3V3\_DIG. REFIO is driven by the external 3.3 V analog reference rail; firmware must select external-reference operation before precision conversions.

## ADC

|       |                                    |           |  |
|-------|------------------------------------|-----------|--|
| Title |                                    | Revision  |  |
| Size  | Number                             | Revision  |  |
| A2    |                                    |           |  |
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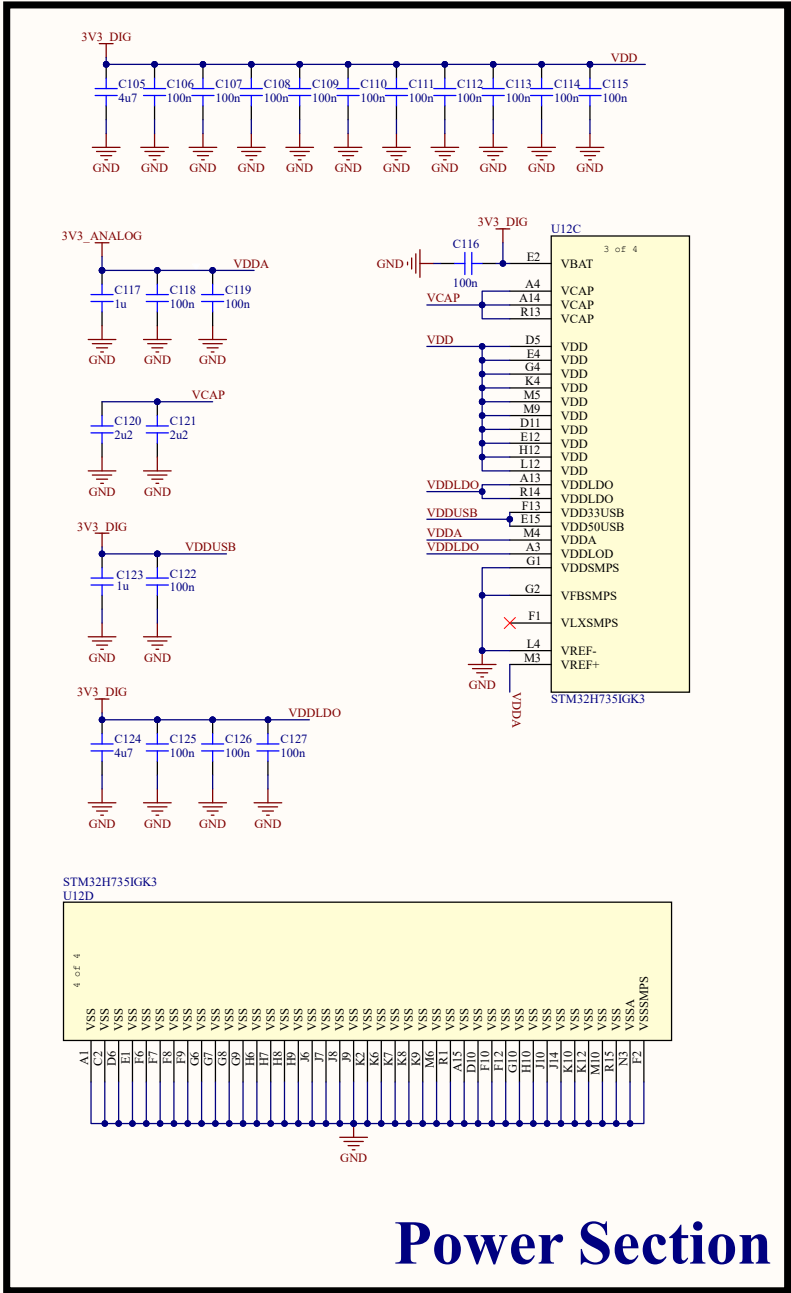
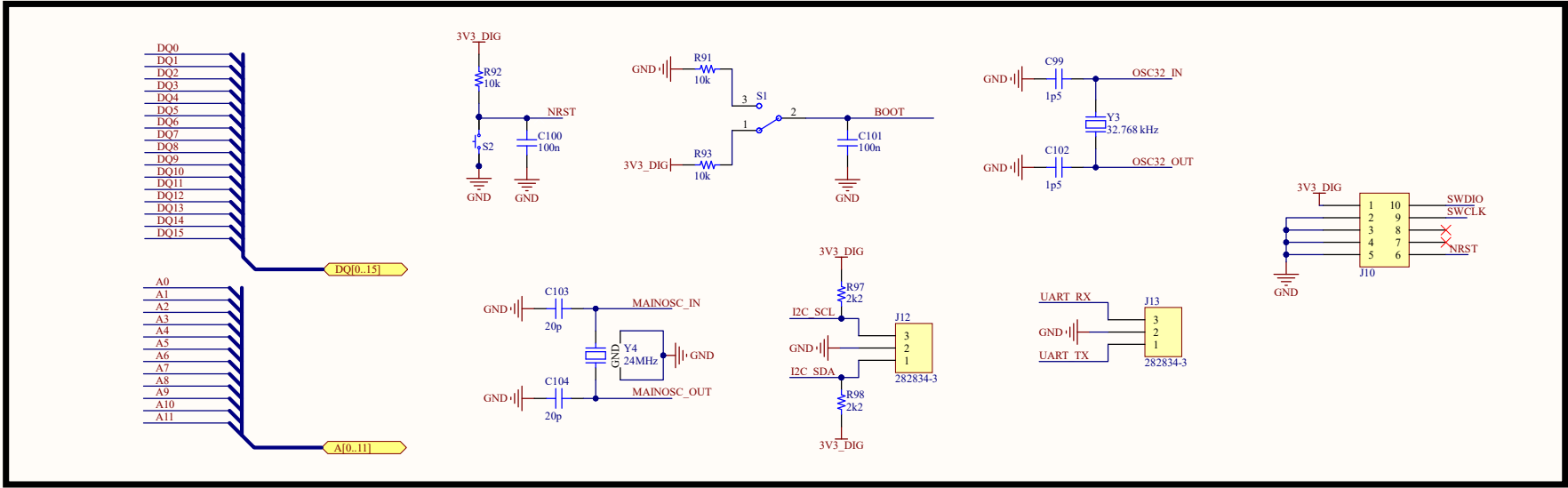
## 4

D

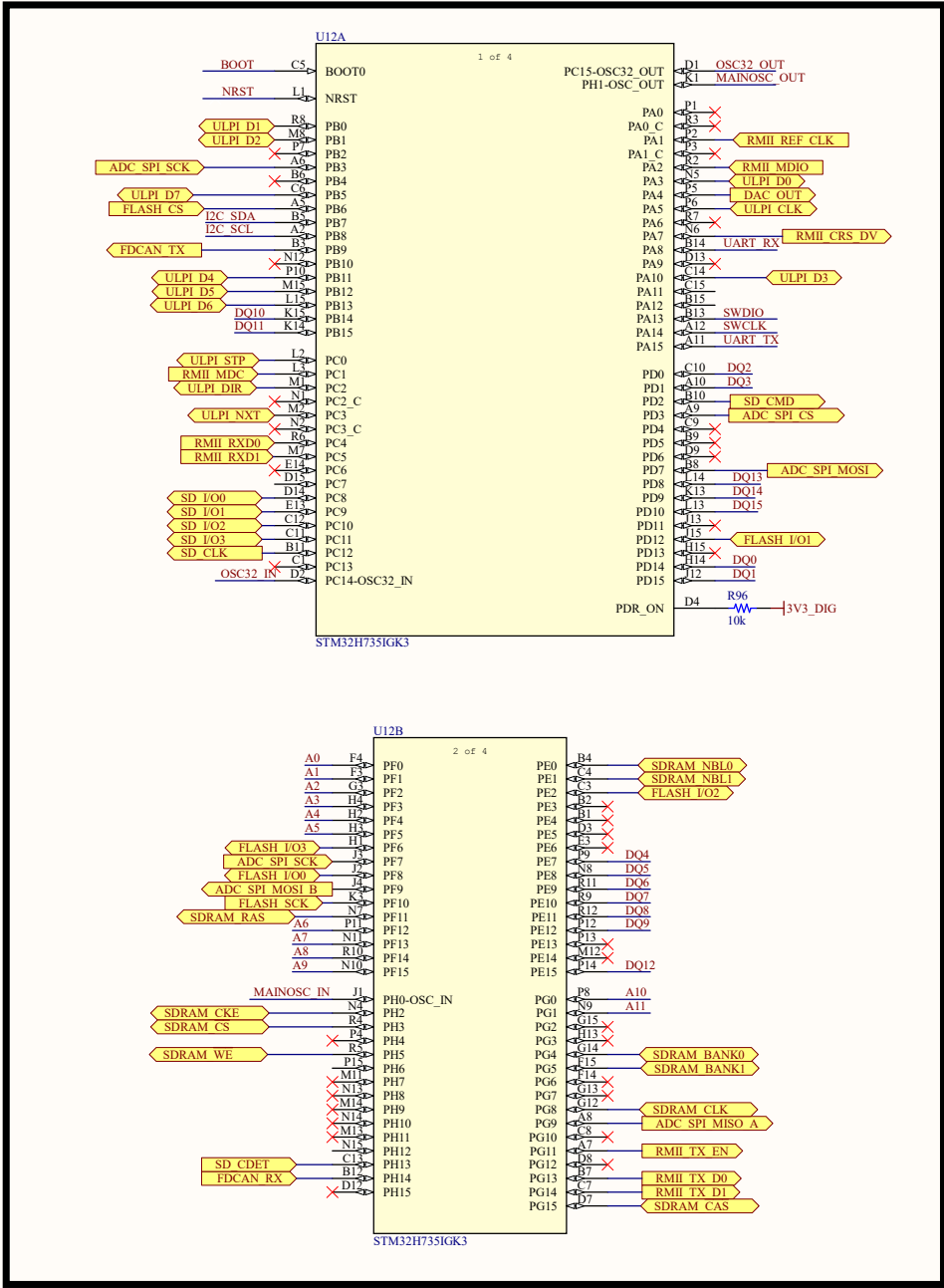
4



# Microcontroller Section



## Power Section



| Title |                                     |           |
|-------|-------------------------------------|-----------|
| Size  | Number                              | Revision  |
| A2    |                                     |           |
| Date: | 9.02.2026                           | Sheet of  |
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